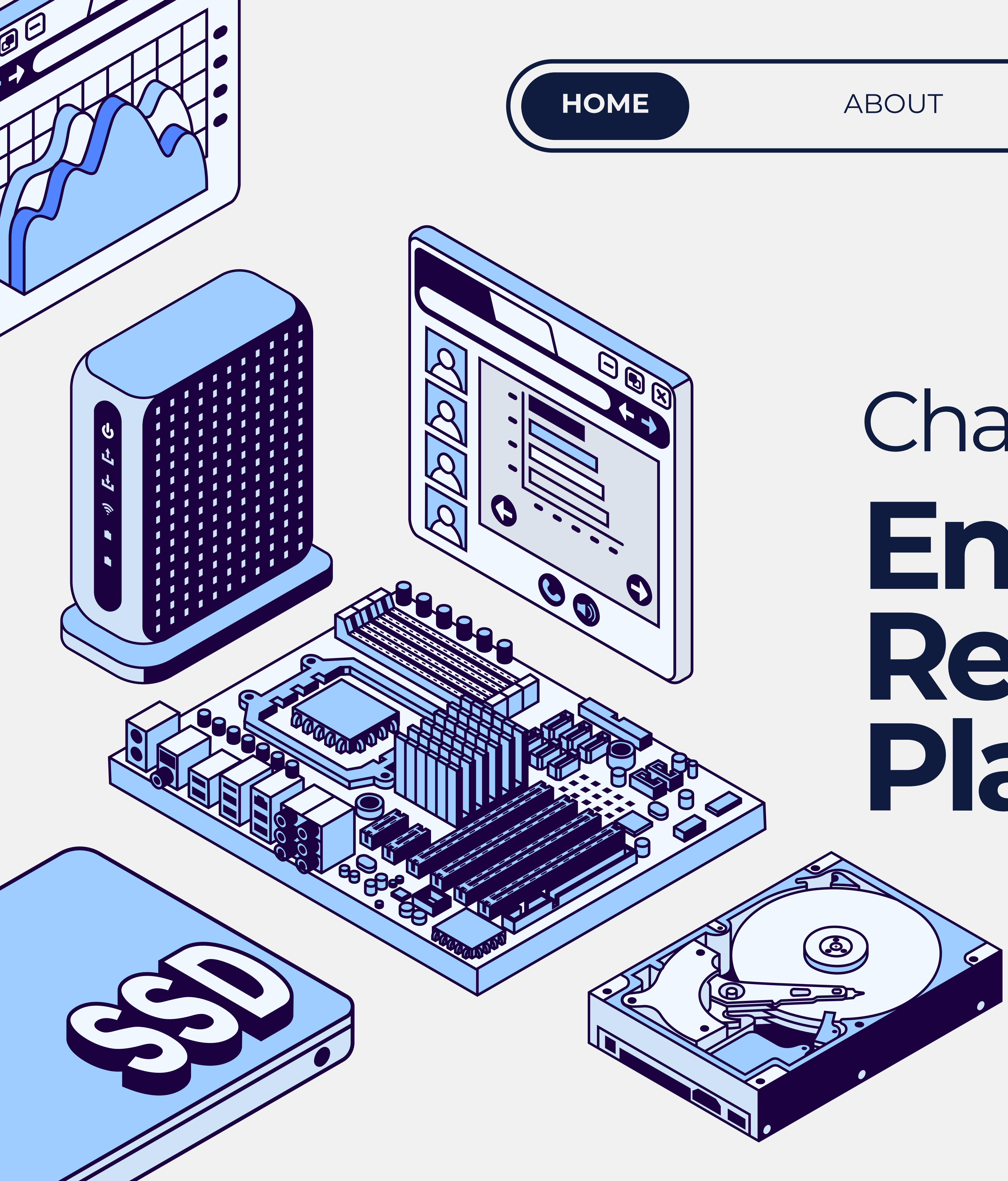


Chapter 11

Enterprise Resource Planning System

Learning Objectives

- Understand the general functionality and key elements of ERP systems.
- Understand the various aspects of ERP configuration including servers, databases, and the use of bolt-on software.
- Understand the purpose of data warehousing as a strategic tool and recognize the issues related to the design, maintenance, and operation of a data warehouse.
- Recognize the risks associated with ERP implementation.
- Be aware of the key considerations related to ERP implementation.
- Understand the internal control and auditing implications associated with ERPs.

GROUP 6

WHAT IS AN ERP?

AN ENTERPRISE RESOURCE PLANNING (ERP) SYSTEM IS A MULTI-MODULE SOFTWARE PACKAGE THAT EVOLVED FROM OLDER SYSTEMS LIKE MRP II (MANUFACTURING RESOURCE PLANNING). THE TERM ERP WAS INTRODUCED BY THE GARTNER GROUP.

Organizations mix and match precoded software components to assemble an enterprise resource planning (ERP) system that best meets their business requirements.

OBJECTIVE OF ERP

INTEGRATE KEY PROCESSES OF THE ORGANIZATION SUCH AS:

1. ORDER ENTRY
2. MANUFACTURING
3. PROCUREMENT
4. ACCOUNTS PAYABLE
5. PAYROLL
6. HUMAN RESOURCES

ERP combines all of these into a single, integrated system that accesses a single database to facilitate the sharing of information and to improve communications across the organization.

TRADITIONAL VS. ERP

FIGURE 11.1 **Traditional Information System**

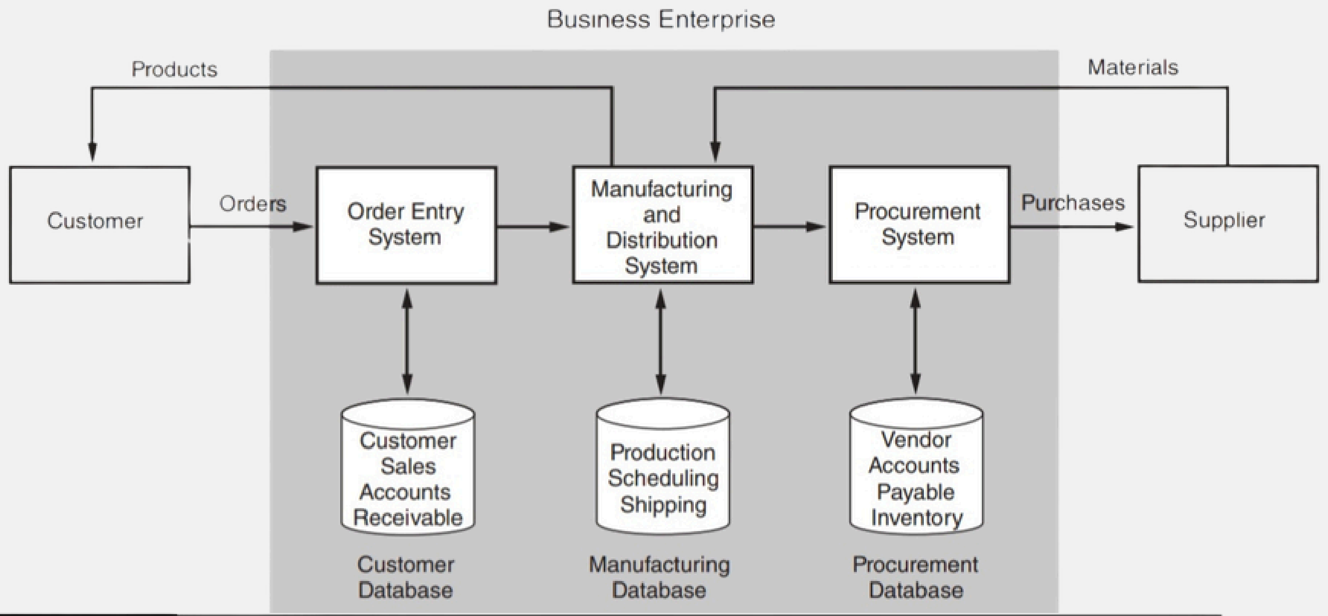
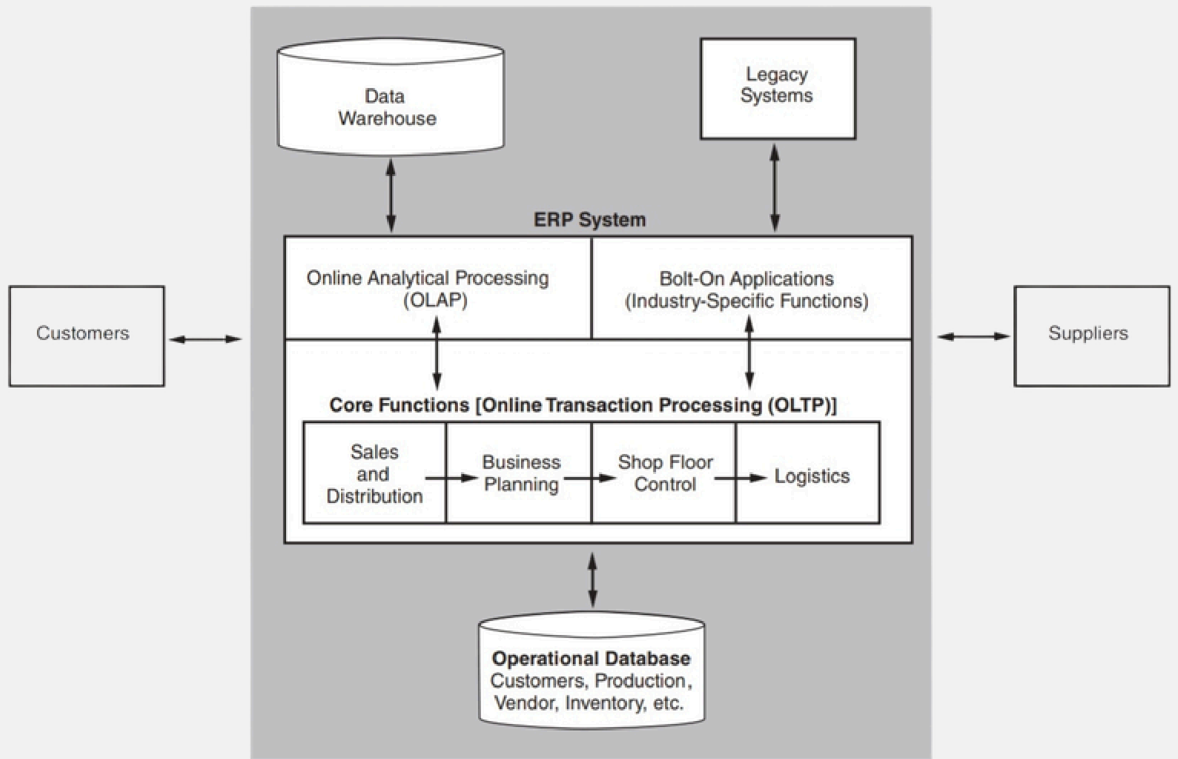


FIGURE 11.2 **ERP System**



ERP Core Applications

ERP Functionality: Core Applications and Business Analysis Application

Core Application (Online Transaction Processing-OLPT)

- applications that operationally support the day-to-day activites of the business.
- if these applications fail, so does the business.
- typical core applications include, but not limited to:
    1. Sales and Distribution
    2. Business Planning
    3. Production Planning
    4. Shop Floor Control
    5. Logistics

Online Analytical Processing (OLAP)

- focuses on analyzing data to help managers make better decisions.
- it includes: decision support, modelling, information retrieval, ad hoc reporting/analysis, and what- if analysis.

ONLINE TRANSACTION PROCESSING (OLPT)

VS.

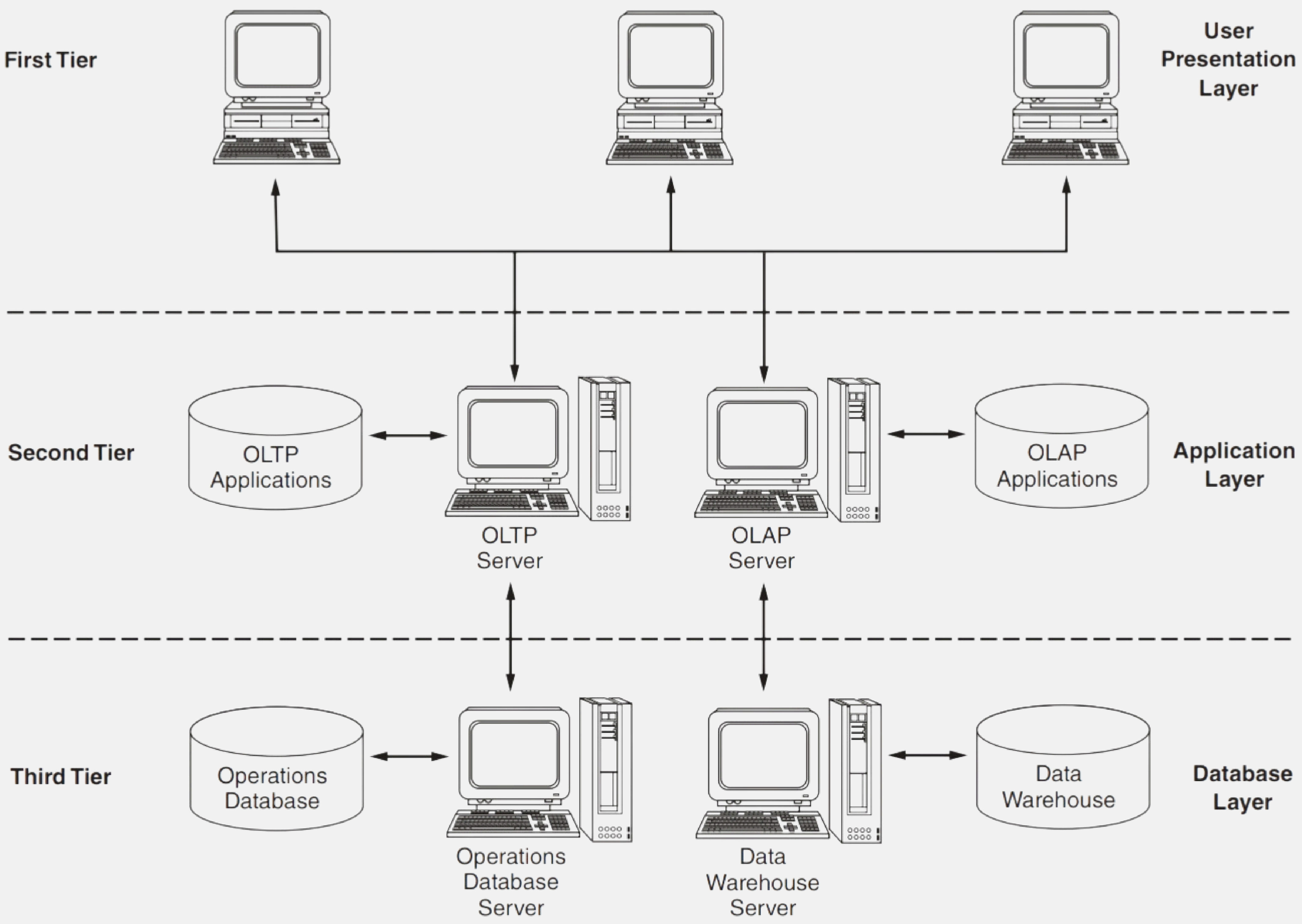
ONLINE ANALYTICAL PROCESSING (OLAP)

ONLINE TRANSACTION PROCESSING (OLPT)	ONLINE ANALYTICAL PROCESSING (OLAP)
Handles **high volume of transactions**	Handles **large amounts of data**
Operational (day-to-day)	Analytical (decision-making)
Simple queries	Complex queries
Current Data	Historical & aggregated data

OLAP servers support common ***analytical operations*** including :

- **Consolidation** : Aggregating data (e.g., sales → region → country)
- **Drill-Down** : Breaking data into detailed levels
- **Slicing and Dicing** : Viewing data from different perspectives (e.g., by product, region, time)

FIGURE 11.5 OLTP and OLAP Client Server



ERP Configuration

1. Server Configuration

- most ERP systems uses the Client-Server Model
- servers may be centralized, while clients are distributed across the enterprise
- two main architectures: 1. Two-Tier Model 2. Three-Tier Model

TWO TIER MODEL

- server performs : application processing and database management
- client computers responsibilities : displaying data to users and sending user input to server
- common in Local Area Network (LAN)
- best suited for small number of users and lower system demand.

THREE TIER MODEL

- separates system function into : Client (user interface), Application server (processing logic), and Database server (data storage)
- process flow : client connects to application server, application server connects to database
- common in Wide Area Netwrok (WAN)
- best suited for large ERP systems and many users and higher processing demands

GROUP 6

HOME ABOUT **CONTENT** OTHERS

FIGURE 11.3 **Two-Tier Client Server**

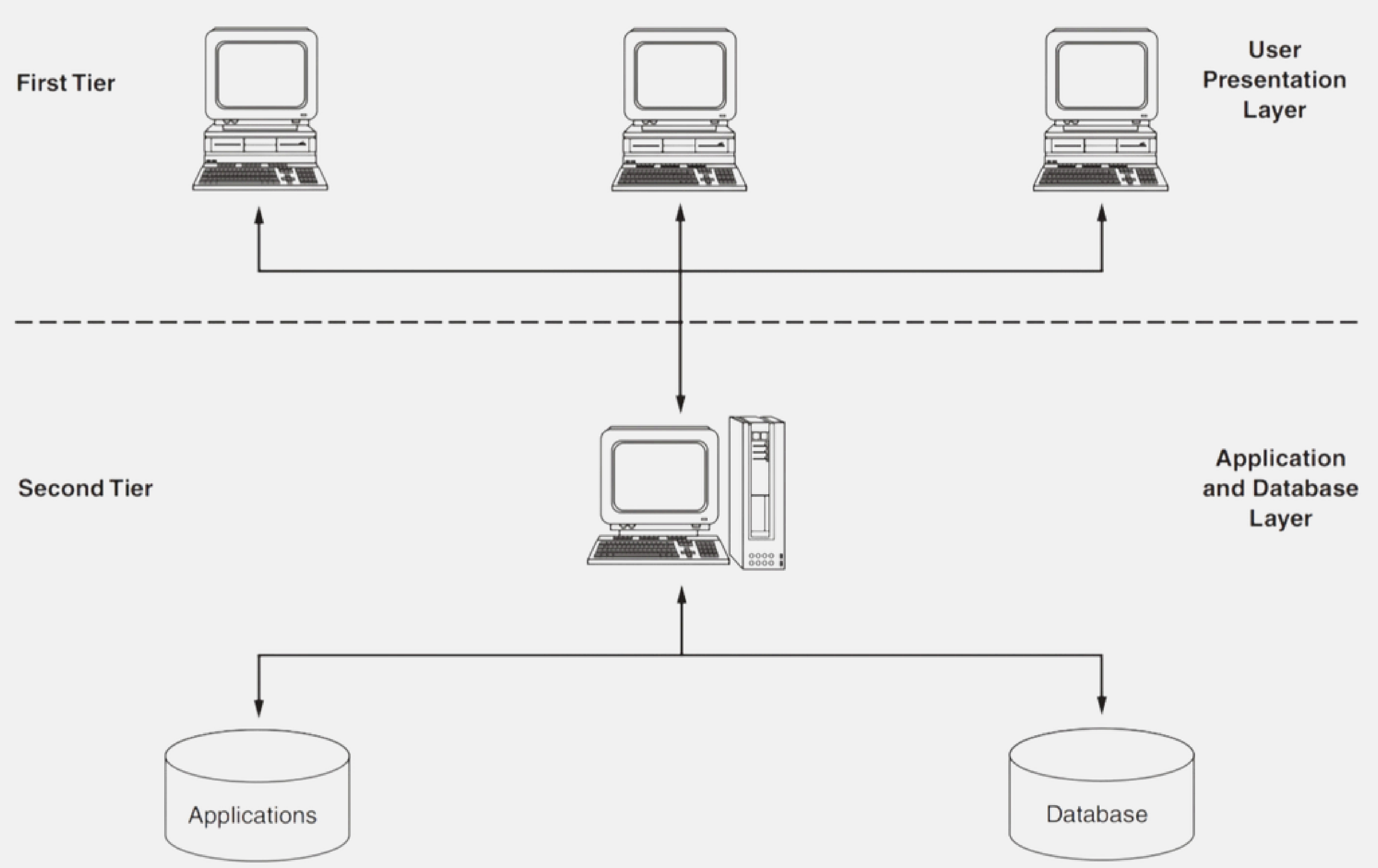
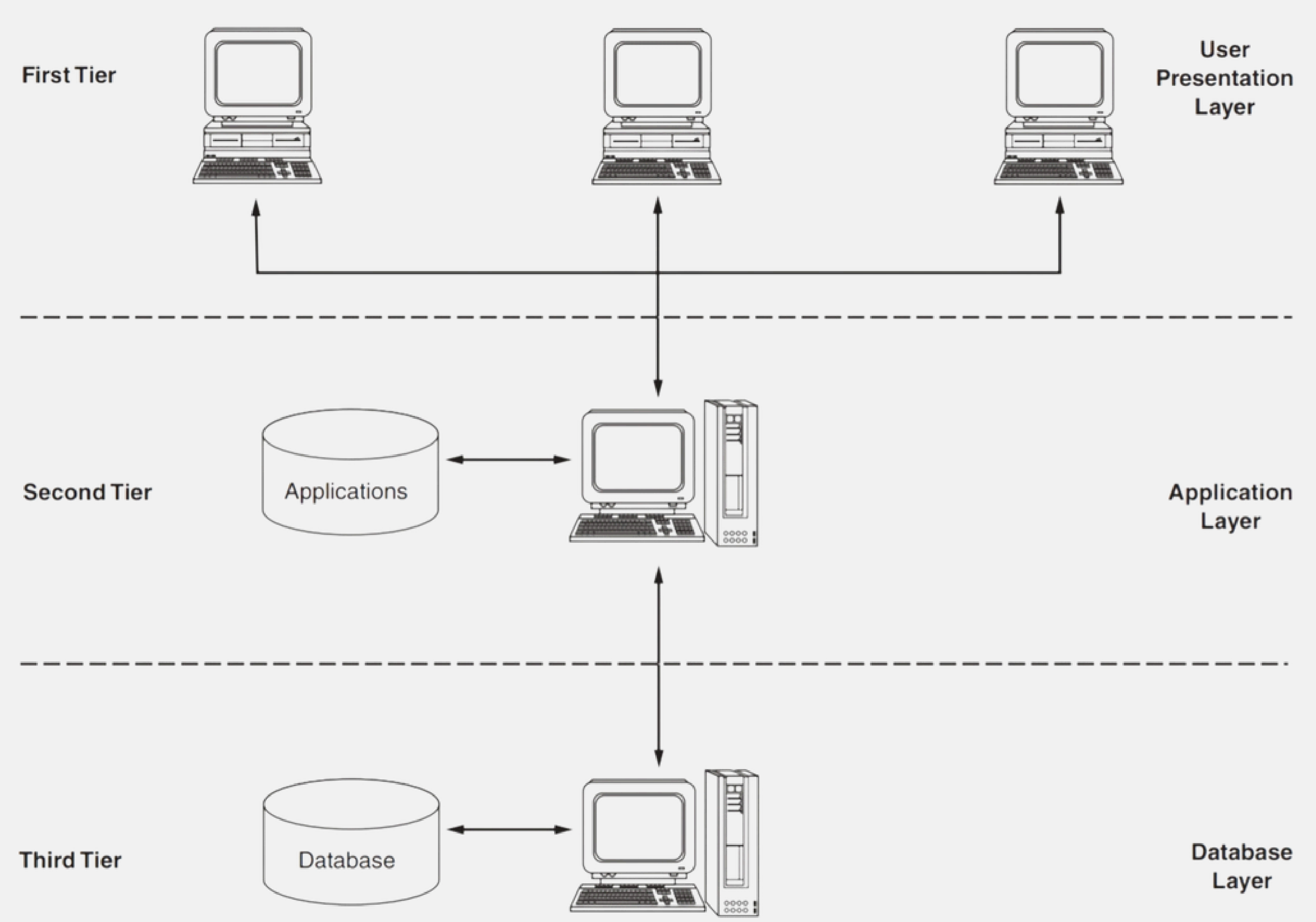


FIGURE 11.4 **Three-Tier Client Server**



2. Database Configuration

- ERP systems contain thousands of database tables, each table is linked to specific business processes
- Configuration is done by: ERP implementation team (users + IT professionals) and Setting system "switches" to select tables and processes
- Requires deep understanding of current business operations
- Often involves: Business Process Reengineering (BPR)
- Companies adapt their processes to fit ERP rather than customizing ERP to fit existing processes

3. Bolt-On Software

- Organizations use bolt-on software to extend functionality
- Additional applications designed to work alongside the ERP system.
- Provided by **third-party vendors** that adds specialized features
- Common reasons for use : Industry-specific needs, advanced or unique processes

Supply Chain Management

- Manage flow of goods from raw materials to final customer
- Key activities : Procurement, Production scheduling, Inventory management, Transportation & warehousing, Customer service & demand forecasting
- Integrate all supply chain processes and connect partners (vendors, logistics provders, etc.)

Data Warehousing

- Large data storage system
- Collects data from different sources
- Used for analysis, not daily transactions

Data Warehouse vs Data Mart

- Warehouse = whole company
- Data Mart = one department

Data Warehousing Process

1. *Modeling data for the data warehouse.*

- is the process of designing how data will be structured in the warehouse. It combines related data to make analysis easier and more efficient compared to complex operational databases.

Data Warehousing Process

2. *Extracting data from operational databases*

- is the process of gathering data from different sources such as databases and files. Only relevant or updated data is usually extracted to improve efficiency and reduce system load.

FIGURE 11.6 **Denormalized Data**

A. Normalized Representation for an Operational Database System

Customer Table

Customer Number	Name	Street	City	State
34675	John Smith	10 Elm	Bath	PA

Invoice Table

Invoice Number	Invoice Date	Shipped Date	Invoice Amount	Customer Number
8866376	06/12/09	06/23/09	600	34675

Line Item Table

Invoice Number	Item Number	Quantity	Price	Extended Price
8866376	j683	2	200	400
8866376	r223	5	40	200

B. Denormalized Representation for Data Warehouse System

Sales Order Table

Customer Number	Name	Street	City	State	Invoice Number	Invoice Date	Shipped Date	Invoice Amount	Item Number	Quantity	Price	Extended Price
34675	John Smith	10 Elm	Bath	PA	8866376	06/12/09	06/23/09	600	j683	2	200	400
34675	John Smith	10 Elm	Bath	PA	8866376	06/12/09	06/23/09	600	r223	5	40	200

Data Warehousing Process

3. *Cleansing extracted data*

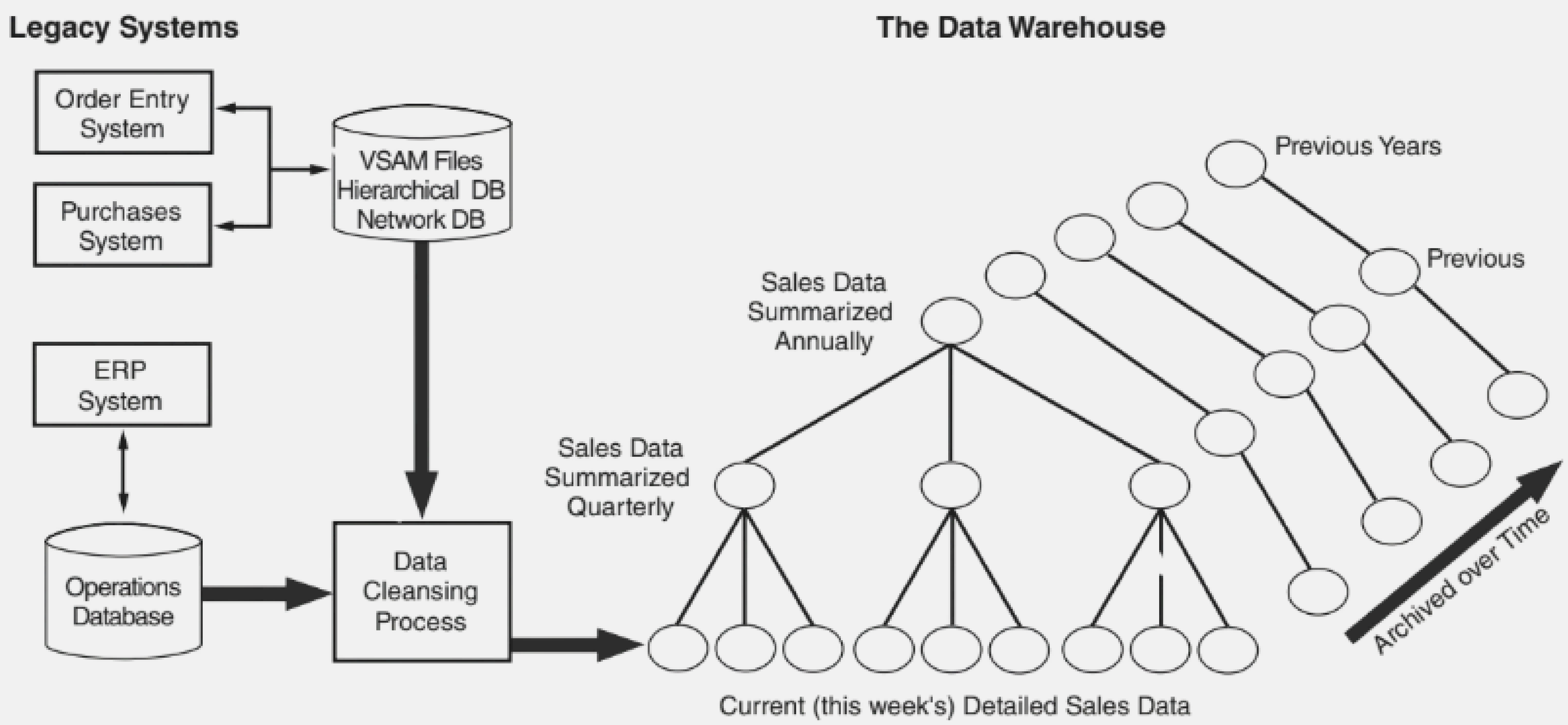
- it involves identifying and correcting errors in the data, such as duplicates, missing values, and incorrect formats. This step ensures that the data used for analysis is accurate and reliable.

Data Warehousing Process

4. *Transforming data into the warehouse model*

- it converts raw data into a more useful format, often by summarizing it into weekly, monthly, or yearly reports. This makes it easier to identify trends and patterns in the data.

FIGURE 11.7 **Data Warehouse System**



Data Warehousing Process

5. *Loading the data into the data warehouse database*

- is the final step where processed data is stored in the data warehouse. The warehouse is kept separate from operational systems to ensure better performance and efficiency.

1. Internal Efficiency
2. Integration of Legacy Systems
3. Consolidation of Global Data

Drill-down Analysis

Drill-down capability is a useful data analysis technique associated with data mining. It begins with the summary views of data described previously. When anomalies or interesting trends are observed, the user drills down to lower-level views and ultimately into the underlying detail data.

RISKS ASSOCIATED WITH ERP IMPLEMENTATIONS

Big Bang vs Phased-in Implementation

Big Bang

- Entire ERP system implemented all at once
- Immediate switch from old system to new system
- Fast implementation
- High risk of failure
- No adjustment period for employees
- Can cause major operational disruptions

Phased-in

- Implemented gradually (step-by-step)
- Done per department or process
- ERP runs with legacy systems temporarily
- Allows adjustment and learning period
- Legacy systems are phased out over time
- Full integration happens gradually

RISKS ASSOCIATED WITH ERP IMPLEMENTATIONS

Opposition to Changes in the Business's Culture

- ERP requires organization-wide involvement
- Needs changes in business processes
- Employees may resist change
- Lack of technical skills and readiness
- Culture may not be open to new systems
- Resistance can lead to delays or failure

Choosing the Wrong ERP

- Goodness of Fit
- System Scalability Issues
  - Size
  - Speed
  - Workload
  - Transaction Cost

RISKS ASSOCIATED WITH ERP IMPLEMENTATIONS

Choosing the Wrong Consultant

- Interview the staff proposed for the project
- Draft a detailed contact
- Establish how staff changes will be handled
- Conduct reference checks
- Align consultant's interests
(pay-for- performance)
- Set a firm termination date

High Cost and Cost Overruns

- ERP has high total cost of ownership (TCO)
- Costs often underestimated
- Training costs higher than expected
- System integration & testing are complex and costly
- Database conversion requires cleaning and validation
- Performance measures need time and resources
- Leads to budget overruns

RISKS ASSOCIATED WITH ERP IMPLEMENTATIONS

Disruptions to Operations

- ERP changes business processes
- Requires adjustment period
- May cause temporary drop in performance
- Leads to delays and inefficiencies
- Risk increases during peak periods
- Big Bang approach can worsen disruptions
- Poor timing = losses in sales and productivity

Implications for Internal Control and Auditing

- **Transaction Authorization**
  - Relies on programmed/automated controls
  - Errors in setup affect entire system
- **Segregation of Duties**
  - Functions are integrated
  - Risk of excess access
  - Requires proper role assignment
- **Supervision**
  - Decision-making shifts to lower levels
  - Managers must understand the system
- **Accounting Records**
  - Faster reporting and data processing
  - Risk of data inaccuracies
- **Independent Verification**
  - Less manual checking
  - Focus on system outputs and performance

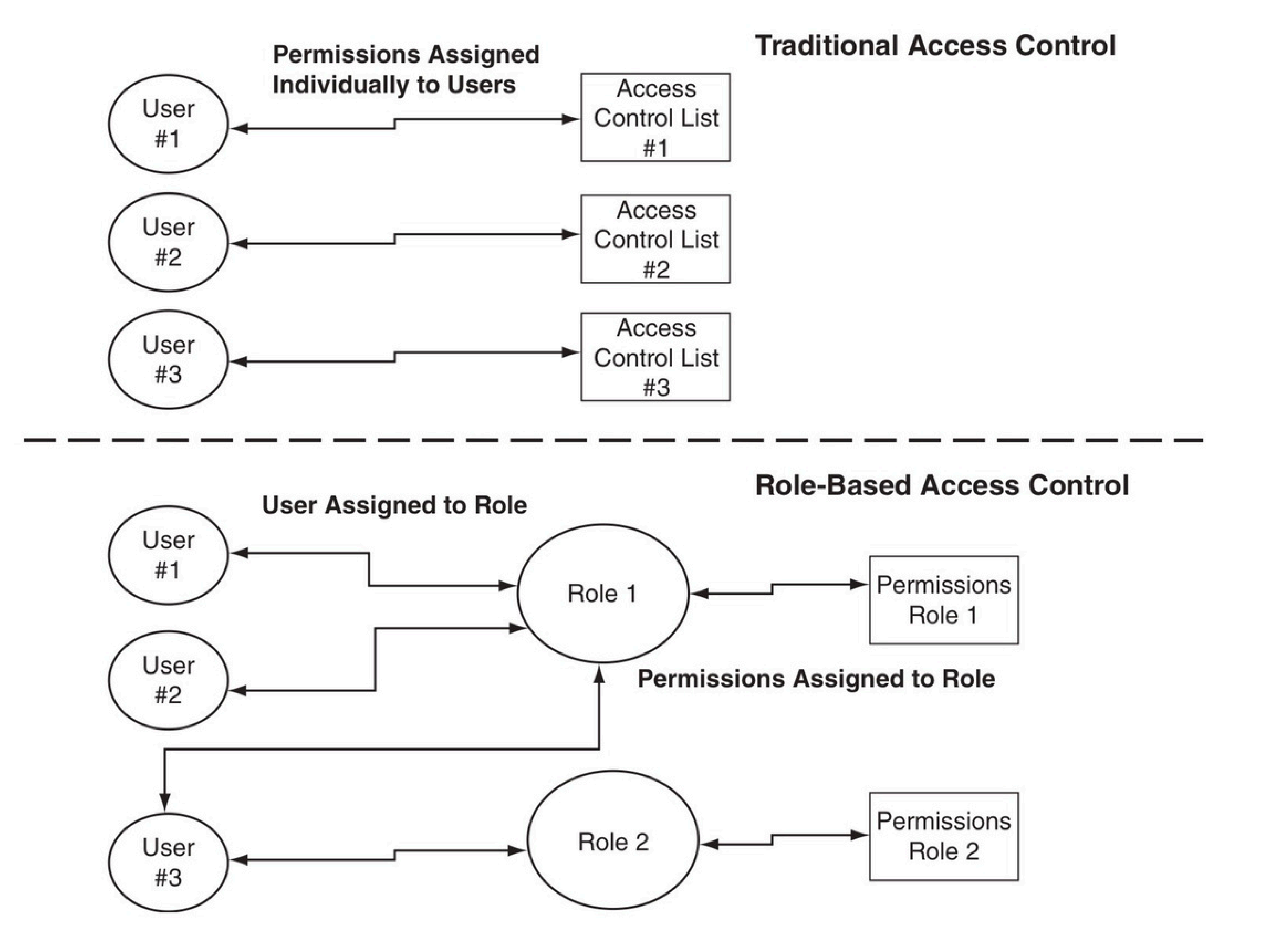
- Access Controls
  - Controls user access and permissions

Access Control Models

- Traditional Model
  - Individual user permissions
  - Harder to manage

- Role-Based Access Control (RBAC)
  - Access based on roles
  - More efficient

FIGURE 11.8 Access Control List versus RBAC



- **SAP Role Definition**
  - **Predefined roles and responsibilities**
  - **Structured access control**

FIGURE 11.9 **SAP Role Definition**

ERP Role Name: **AP Processing** **Location: AP Department**

Role Goal: Data Entry of Customer Invoices and Check Requests

Knowledge, Skills: Knowledge of invoice processing from receipt to payment and relevant finance and procurement SAP modules, Vendor Master Data, P-card processes. Understanding of IRS 1099 reporting & processing requirements.

Responsibilities/Tasks:

--Review and verify parked invoices and submit to Accounts Payable
--Park non-P.O. invoices (check request)
--Enter invoices for goods received into SAP; code invoices for payment (park)

SAP Security Considerations:

AP	**Transaction Type**		
FB03	Display Document	FB04	Display Changes
FBD3	Display Recurring Entry	FBL1	
FBL1N	Display Vendor Line Item	FBL3N	Account Line Item Display
FBV2	Change Parked Document	FBV3	Display Parked Documents
FCH1	Display Check Information	FCH2	Display Payment Document checks
FCHN	Check Register	FK10N	Vendor Balance Display
FV60	Park/Edit Invoice	PV65	Park/Edit Credit Memo
MIR4	Display Invoice Document	MIR5	Display List of Invoice

INTERNAL CONTROL ISSUES RELATED TO ERP CONTROLS

1. Creation of Unnecessary Roles
2. The rule of least access should apply to permission assignments
3. Monitor role creation and permission-granting activities

Contingency Planning

- ERP systems can be risky since everything depends on one system
- Two general approaches to mitigate this risk:

  For:

  1. Centralized Systems
  2. Decentralized Systems

LEADING ERP PRODUCTS

- SAP
- ORACLE
- MICROSOFT
- SOFTBRANDS

SAP

mySAP ERP Financials

- Financial and managerial accounting
- Corporate Governance
- Financial Supply Chain Management

mySAP ERP Operations

- Procurement and Logistics Execution
- Product Development and Manufacturing
- Sales and Services

mySAP ERP Human Capital Management

- Workforce Process Management
- Workforce Deployment

mySAP ERP Corporate Services

- Centralized and Decentralized administrative processes

ORACLE | PEOPLESOFT

- Oracle E-Business Suite
- PeopleSoft Enterprise Solutions
- J.D. Edwards EnterpriseOne

MICROSOFT

Microsoft Dynamics GP

Microsoft Dynamics AX

Microsoft Dynamics SL

SAGE SOFTWARE

- Manufacturing
- Distribution
- Project Accounting

SOFTBRANDS

Customs Product Manufacturing

Trace and Serialization

THANK YOU!